

Federico Mazzone

Cork, Ireland · mazzoni.federico1@gmail.com · [LinkedIn Profile](#) · [GitHub Portfolio](#) · [Scholar](#)

Built my company's production-ready AI systems, including a customer-facing LLM chatbot serving 100+ daily users and internal AI automation tools, accelerating workflows from days to hours. Experience in NLP, ML, AWS. 8 published papers.

EXPERIENCE

AI Engineer

Domotz

May 2024 – Present

Dublin, Ireland (Remotely)

- **Built Domotz AI Stack:** Established AI roadmap and workflows, enabling rapid prototyping and deployment.
- **Chatbot:** Engineered a FastAPI RAG-based chatbot (using Qdrant, LangChain, PostgreSQL) deployed on AWS, boosting customer engagement by 65% and serving 100+ daily users. Built multiple knowledge bases and UIs.
- **Dynamic MCP:** Developed an MCP server acting as a semantic proxy across multiple MCP servers, dynamically providing only the tools and skills relevant to each user's task.
- **Conversation Intelligence:** Engineered an in-house Gong alternative for Sales & CS teams, featuring zero-shot LLM classification, summarization, and RAG-based chat over thousands of emails, tickets, and call transcripts.
- **AI Automation Systems:** Built LLM-powered automation systems that reduced sales, finance, and marketing workflows from days to hours, including a Slack sales bot and an automated invoice processor.
- **Anomaly Detection:** Combined traditional ML and LLM reasoning to detect network anomalies in time series.
- Coordinated with multiple stakeholders, and leveraged platforms such as AWS, BitBucket, Jira, Zendesk, n8n.

PhD Research Fellow in AI

University of Pisa / National Research Council of Italy

Nov. 2022 – May 2024

Pisa, Italy

- **Human-in-the-Loop:** Built a hybrid AI-human feedback loop system, boosting label accuracy up to 50%.
- **NLP:** Developed a library to quantify trust in social networks using embedding similarity and information density.
- **Algorithmic Fairness:** Designed algorithms to improve fairness in social networks and tabular ML models.
- Published 8 peer-reviewed articles, managed multiple concurrent projects from concept to experimental validation, gained expertise in analysing complex data, and presented research at conferences.
- Conducted 1.5 years of PhD research, before transitioning to industry for practical applications of AI.

Freelance Data Analyst

Self-Employed

2022

Pisa, Italy

- **Recommendation System:** Developed content-based and collaborative filtering recommendation systems.
- **Data Collection:** Created datasets by scraping Wikipedia and independent sites with Selenium/Beautiful Soup.
- **Bioinformatics:** Explored methods in *R* to identify potential biomarkers in Rett syndrome data.

PERSONAL PROJECTS

StockAgent: Multi-agent AutoGen system for stock market analysis with a custom MCP yfinance server.

HINTT: Real-time OCR+translation tool sending screenshots to OpenAI/Ollama. Employed tkinter and mss.

Simple Palette Maker: An app to colorize black-and-white GameBoy palettes. Deployed on Streamlit. 1k+ users.

SKILLS

AI/ML Stack: LLMs (LangChain, LangGraph, AutoGen, Bedrock), RAG, HuggingFace, SK-Learn, PyTorch, RiverML

Backend & Deployment: FastAPI, Streamlit, Docker, AWS (EC2, S3, CloudWatch, Terraform), PostgreSQL, Qdrant

Programming: Python, JavaScript, R

Platforms: Git, Jira, BitBucket, Snowflake, Zendesk, Salesforce, n8n, OpenAI, Claude, Amazon Bedrock

Languages: Italian (Native), English (C1), German (A2)

EDUCATION

University of Southern Denmark

Summer School on Deep Learning

2023

Odense, Denmark

University of Pisa

Master's Degree in Digital Humanities (Data Science and NLP) — 110/110 with honours

2020 – 2022

Pisa, Italy

- Relevant Courses: Data Mining, Social Network Analysis, Text Analytics, Distributed Data Analysis and Mining, Semantic Web, Laboratory of Data Science, Applied Linguistics, Computational Linguistics.

University of Pisa

Bachelor's Degree in Philosophy — 110/110 with honours

2014 – 2017

Pisa, Italy